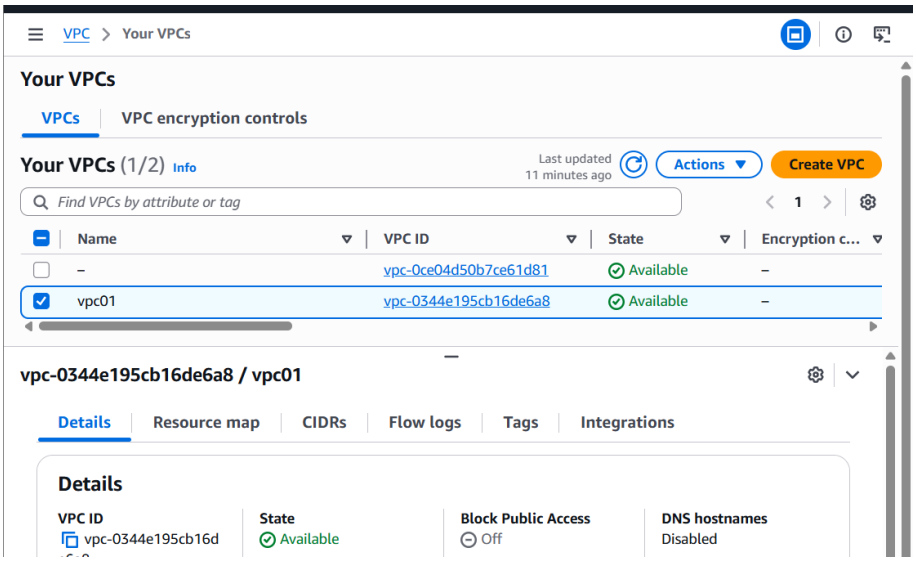
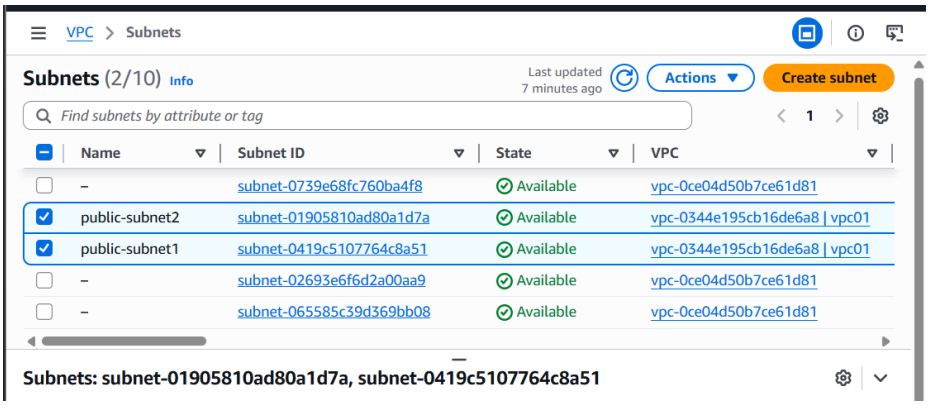
Capstone Project: Enterprise AWS Infrastructure Deployment

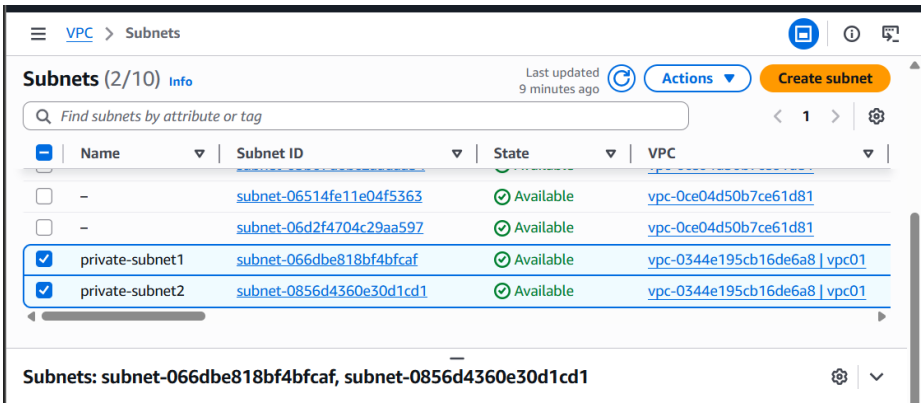
2. Networking Layer (VPC Design)

VPC:

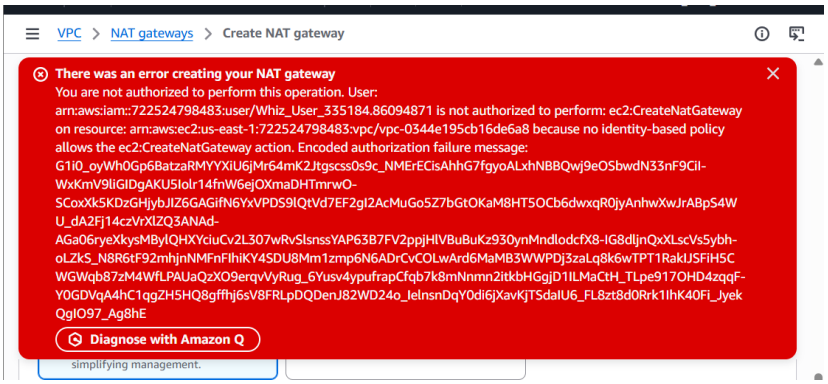


Subnets:

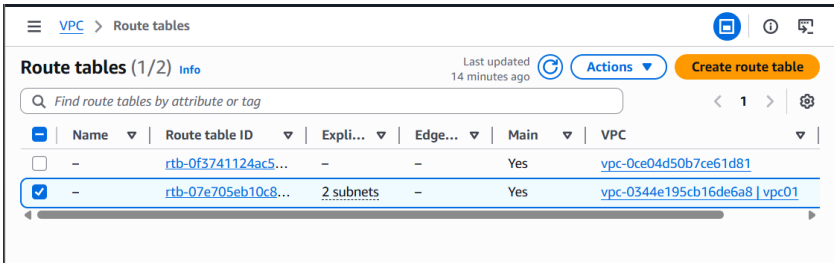


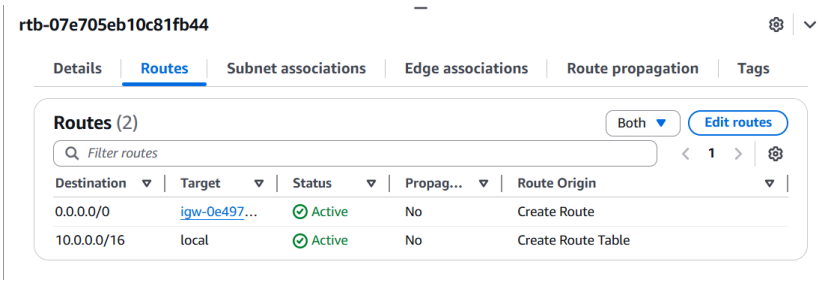


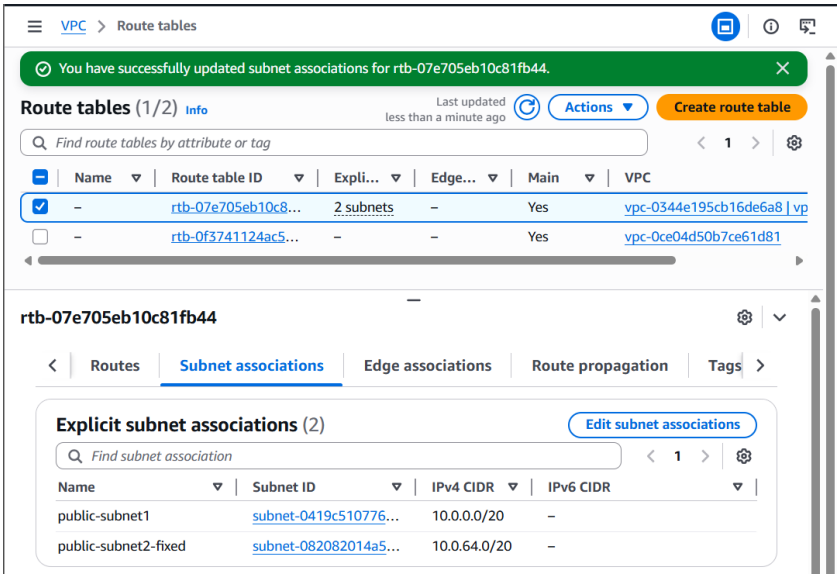
NAT Gateway:



Route Table:

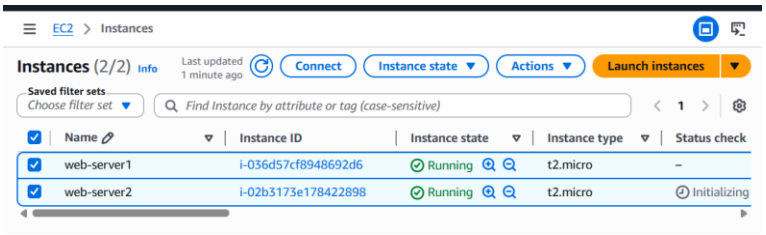


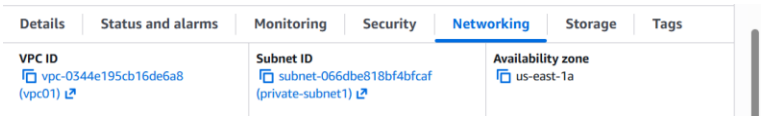


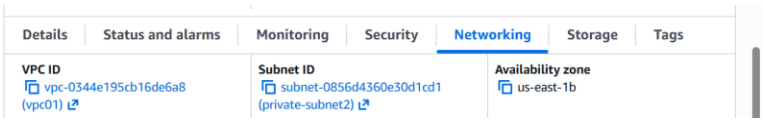


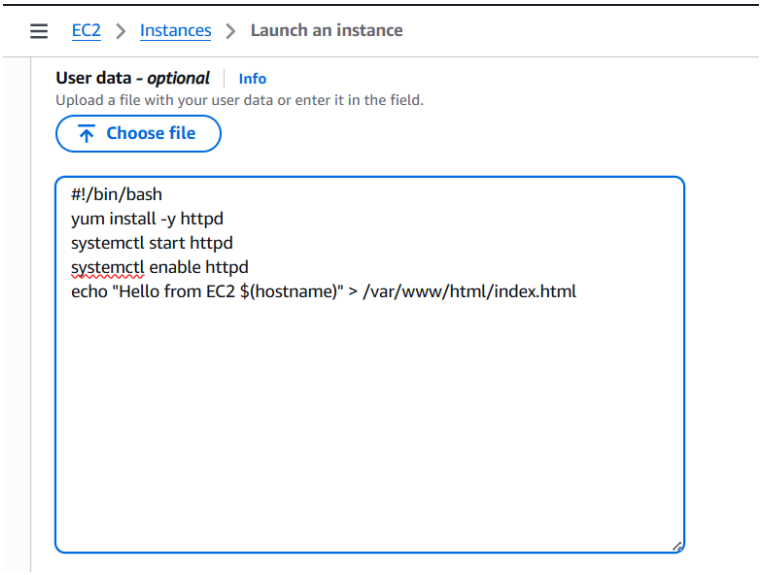
3. Compute Layer

EC2 Instances in private subnets with web server (Apache) installed:

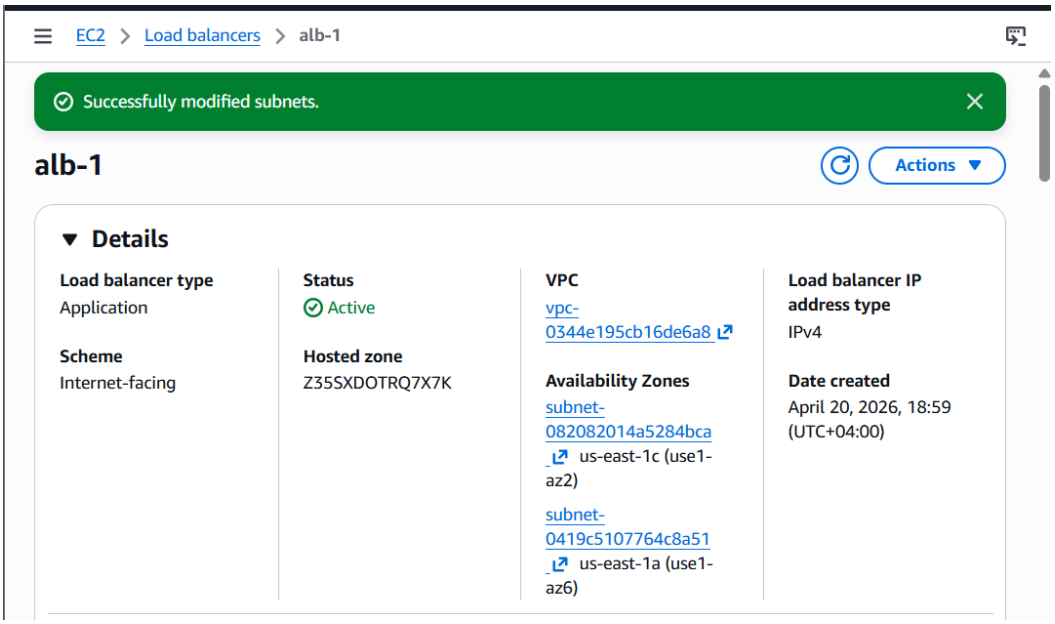


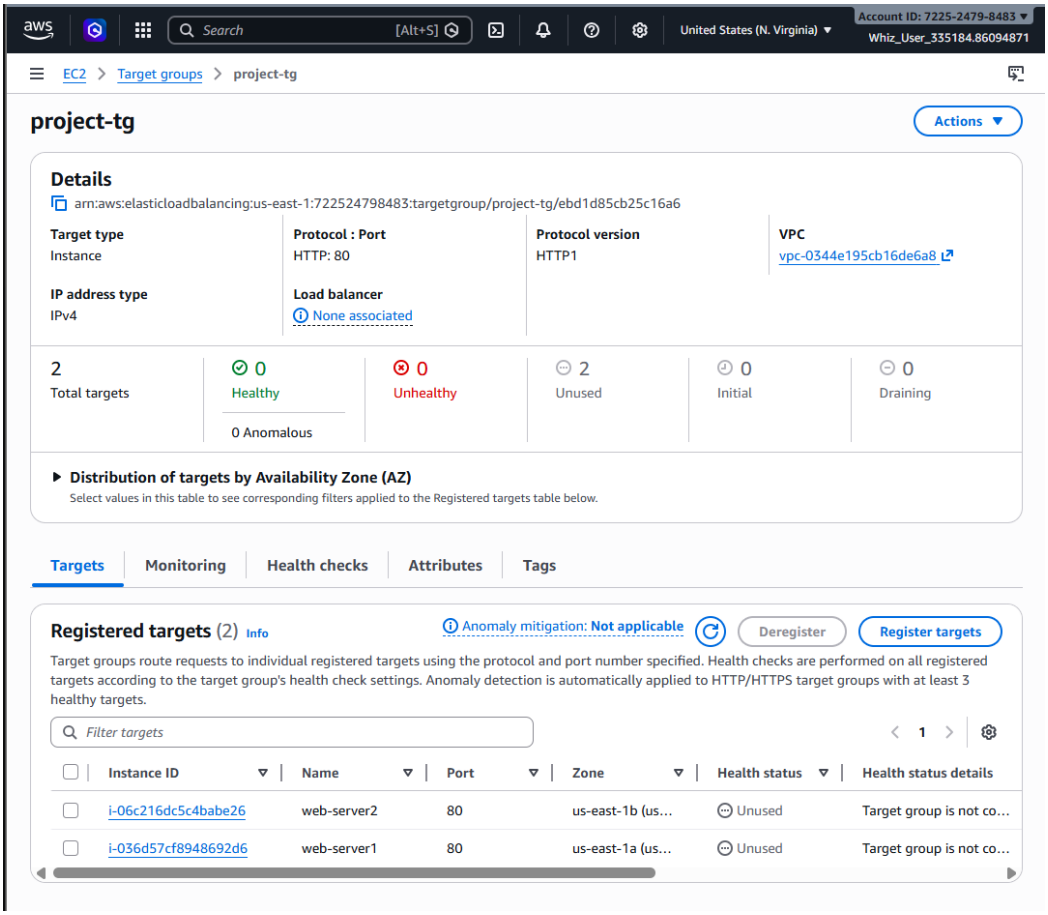






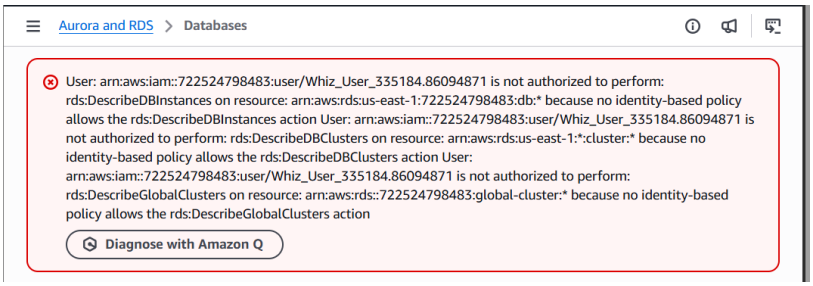
4. Load Balancing:





5. Data Layer

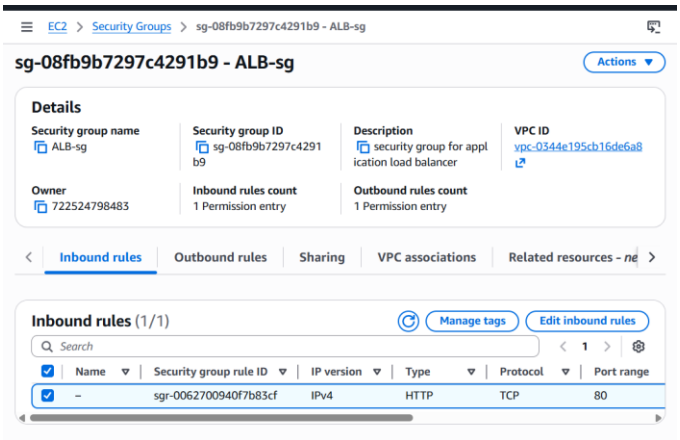
Due to sandbox restrictions



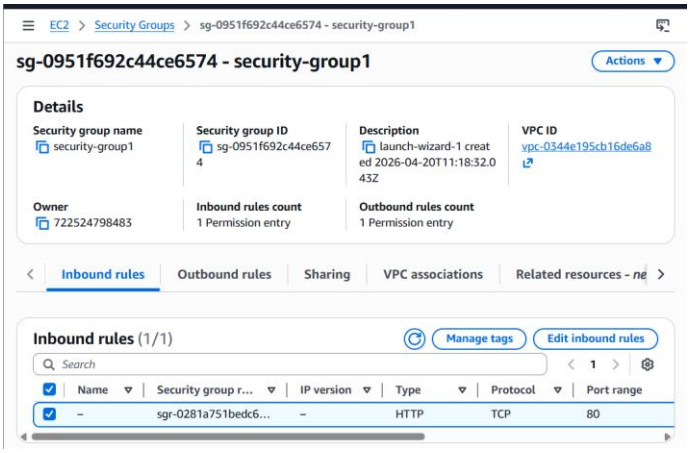
6. Security

Security Groups

Application load balancer SG:

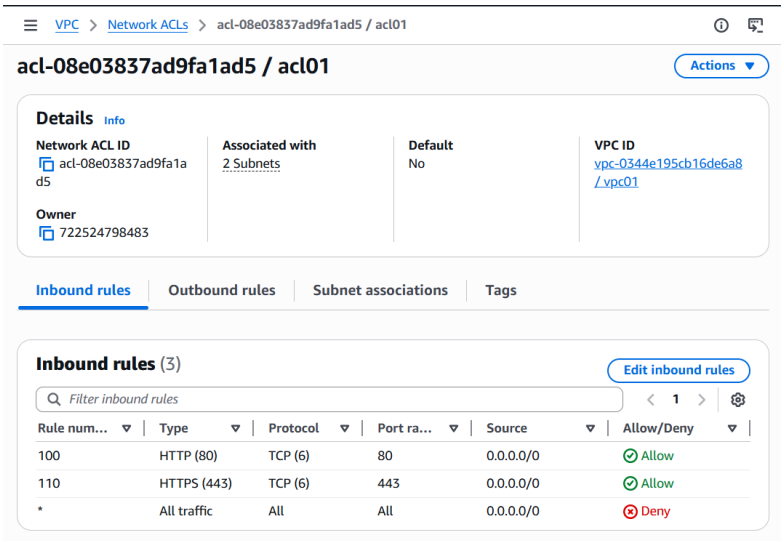


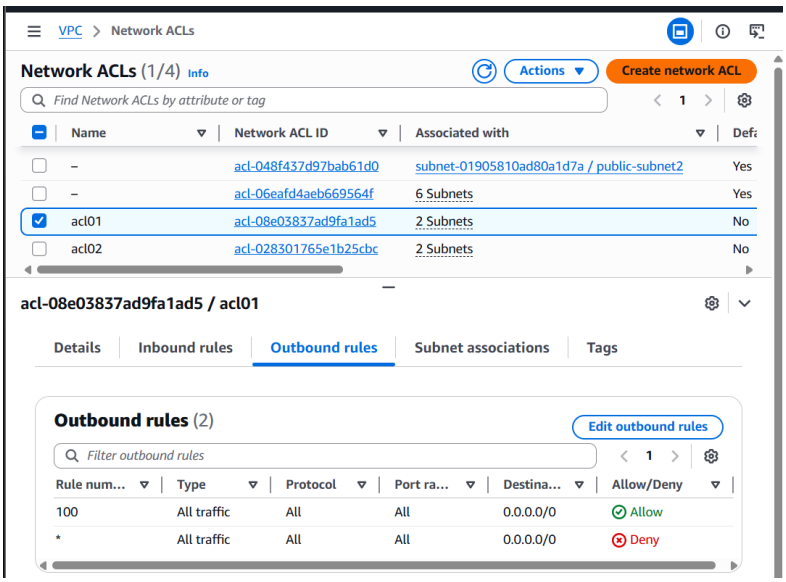
EC2 SG:

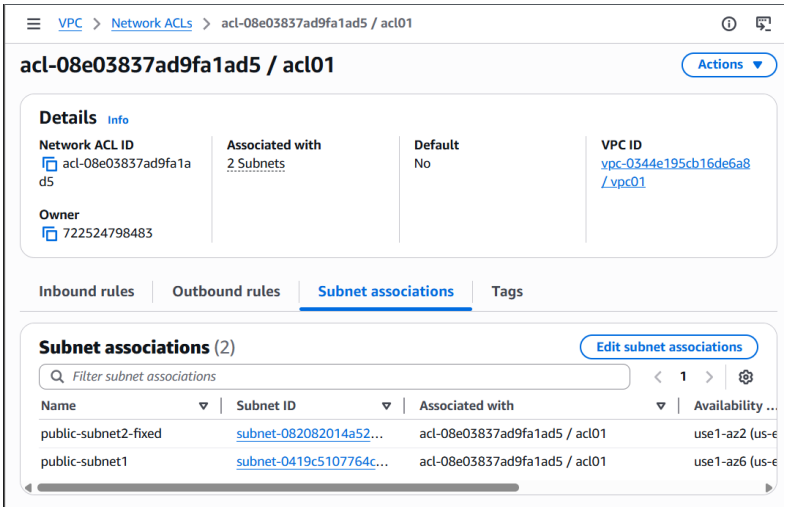


NACL

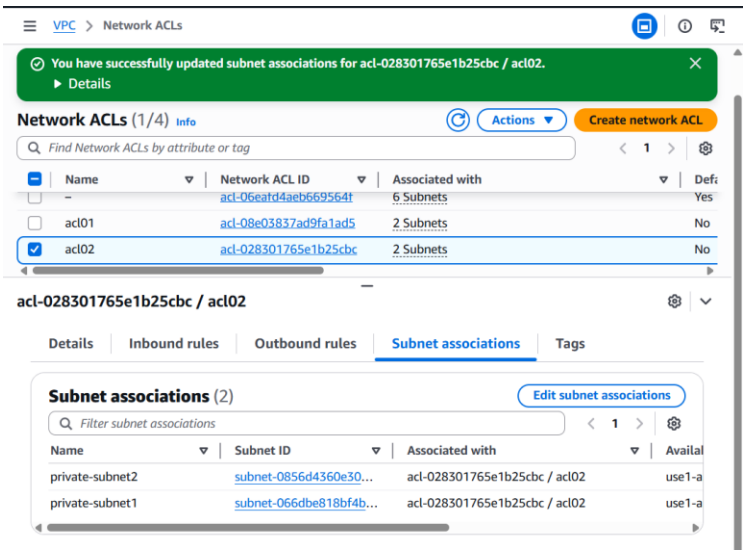
For LB:

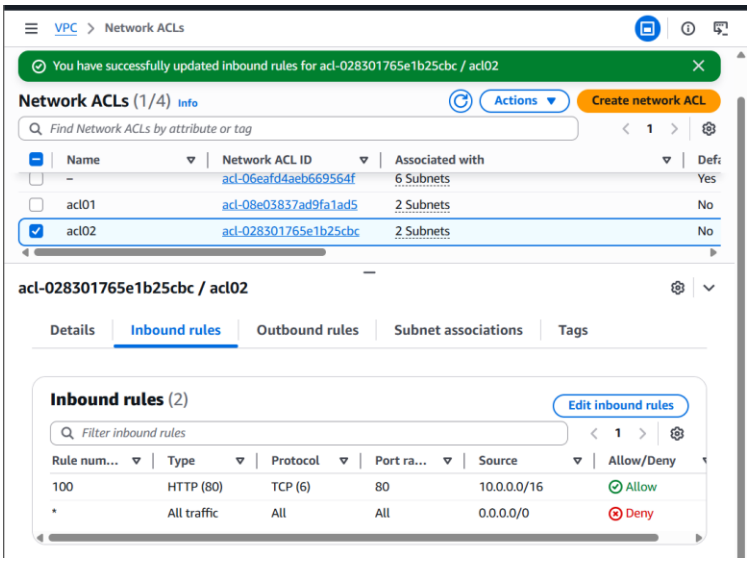

VPC > Network ACLs > acl-08e03837ad9fa1ad5 / acl01
acl-08e03837ad9fa1ad5 / acl01
Actions
Details Info
Network ACL ID
acl-08e03837ad9fa1ad5
Associated with
2 Subnets
Default
No
VPC ID
vpc-0344e195cb16de6a8 / vpc01
Owner
722524798483
Inbound rules
Outbound rules
Subnet associations
Tags
Inbound rules (3)
Edit inbound rules
Filter inbound rules
Rule num... | Type | Protocol | Port ra... | Source | Allow/Deny
100 | HTTP (80) | TCP (6) | 80 | 0.0.0.0/0 | Allow
110 | HTTPS (443) | TCP (6) | 443 | 0.0.0.0/0 | Allow
* | All traffic | All | All | 0.0.0.0/0 | Deny

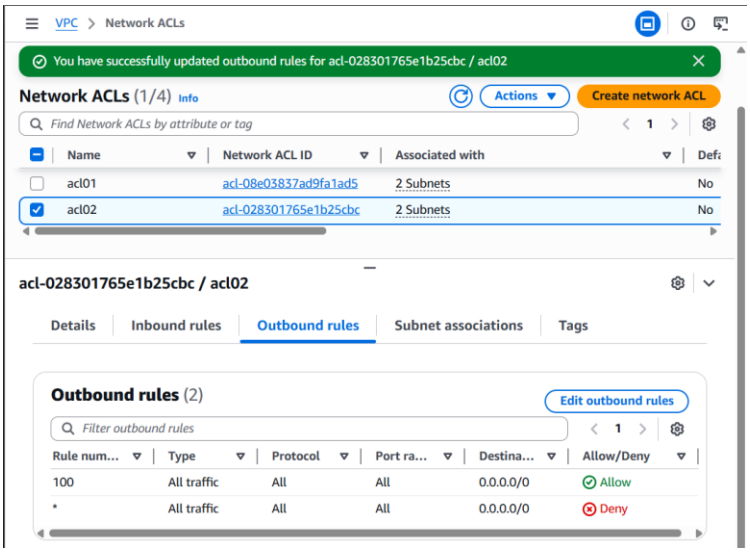

VPC > Network ACLs
Network ACLs (1/4) Info
Actions
Create network ACL
Find Network ACLs by attribute or tag
Name | Network ACL ID | Associated with | Defa
– | acl-048f437d97bab61d0 | subnet-01905810ad80a1d7a / public-subnet2 | Yes
– | acl-06eafd4aeb669564f | 6 Subnets | Yes
acl01 | acl-08e03837ad9fa1ad5 | 2 Subnets | No
acl02 | acl-028301765e1b25cbc | 2 Subnets | No
acl-08e03837ad9fa1ad5 / acl01
Details
Inbound rules
Outbound rules
Subnet associations
Tags
Outbound rules (2)
Edit outbound rules
Filter outbound rules
Rule num... | Type | Protocol | Port ra... | Destina... | Allow/Deny
100 | All traffic | All | All | 0.0.0.0/0 | Allow
* | All traffic | All | All | 0.0.0.0/0 | Deny


VPC > Network ACLs > acl-08e03837ad9fa1ad5 / acl01
acl-08e03837ad9fa1ad5 / acl01
Actions
Details Info
Network ACL ID
acl-08e03837ad9fa1ad5
Associated with
2 Subnets
Default
No
VPC ID
vpc-0344e195cb16de6a8 / vpc01
Owner
722524798483
Inbound rules
Outbound rules
Subnet associations
Tags
Subnet associations (2)
Edit subnet associations
Filter subnet associations
Name
Subnet ID
Associated with
Availability ...
public-subnet2-fixed
subnet-082082014a52...
acl-08e03837ad9fa1ad5 / acl01
use1-az2 (us-e
public-subnet1
subnet-0419c5107764c...
acl-08e03837ad9fa1ad5 / acl01
use1-az6 (us-e

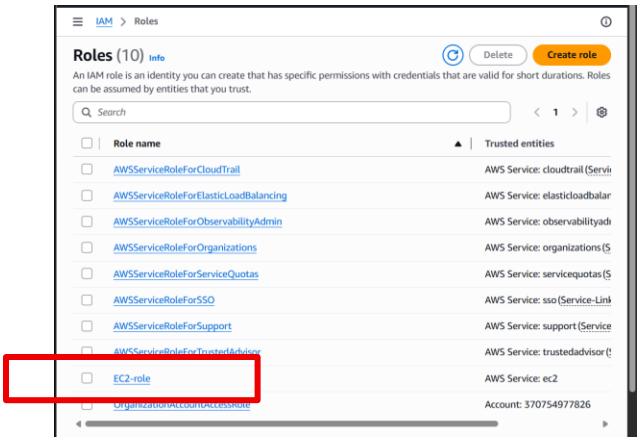
For EC2:


VPC > Network ACLs
You have successfully updated subnet associations for acl-028301765e1b25cbc / acl02.
Details
Network ACLs (1/4) Info
Actions
Create network ACL
Find Network ACLs by attribute or tag
Name
Network ACL ID
Associated with
acl-06eafd4aeb669564f
6 Subnets
Yes
acl01
acl-08e03837ad9fa1ad5
2 Subnets
No
acl02
acl-028301765e1b25cbc
2 Subnets
No
acl-028301765e1b25cbc / acl02
Details
Inbound rules
Outbound rules
Subnet associations
Tags
Subnet associations (2)
Edit subnet associations
Filter subnet associations
Name
Subnet ID
Associated with
private-subnet2
subnet-0856d4360e30...
acl-028301765e1b25cbc / acl02
private-subnet1
subnet-066dbe818bf4b...
acl-028301765e1b25cbc / acl02

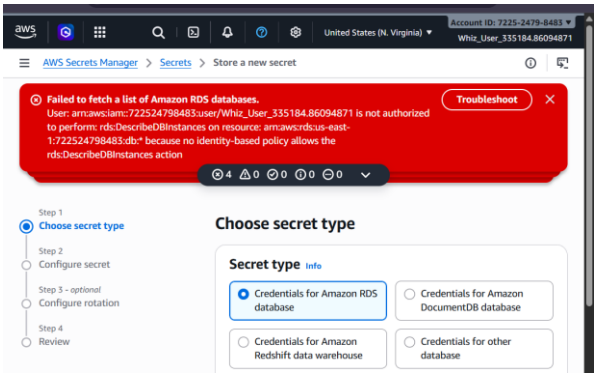

VPC > Network ACLs
You have successfully updated inbound rules for acl-028301765e1b25cbc / acl02
Network ACLs (1/4) Info
Actions
Create network ACL
Find Network ACLs by attribute or tag
Name
Network ACL ID
Associated with
Defa
–
acl-06eafd4aeb669564f
6 Subnets
Yes
acl01
acl-08e03837ad9fa1ad5
2 Subnets
No
acl02
acl-028301765e1b25cbc
2 Subnets
No
acl-028301765e1b25cbc / acl02
Details
Inbound rules
Outbound rules
Subnet associations
Tags
Inbound rules (2)
Edit inbound rules
Filter inbound rules
Rule num...
Type
Protocol
Port ra...
Source
Allow/Deny
100
HTTP (80)
TCP (6)
80
10.0.0.0/16
Allow
*
All traffic
All
All
0.0.0.0/0
Deny


VPC > Network ACLs
You have successfully updated outbound rules for acl-028301765e1b25cbc / acl02
Network ACLs (1/4) Info
Actions
Create network ACL
Find Network ACLs by attribute or tag
Name
Network ACL ID
Associated with
Defa
acl01
acl-08e03837ad9fa1ad5
2 Subnets
No
acl02
acl-028301765e1b25cbc
2 Subnets
No
acl-028301765e1b25cbc / acl02
Details
Inbound rules
Outbound rules
Subnet associations
Tags
Outbound rules (2)
Edit outbound rules
Filter outbound rules
Rule num...
Type
Protocol
Port ra...
Destina...
Allow/Deny
100
All traffic
All
All
0.0.0.0/0
Allow
*
All traffic
All
All
0.0.0.0/0
Deny

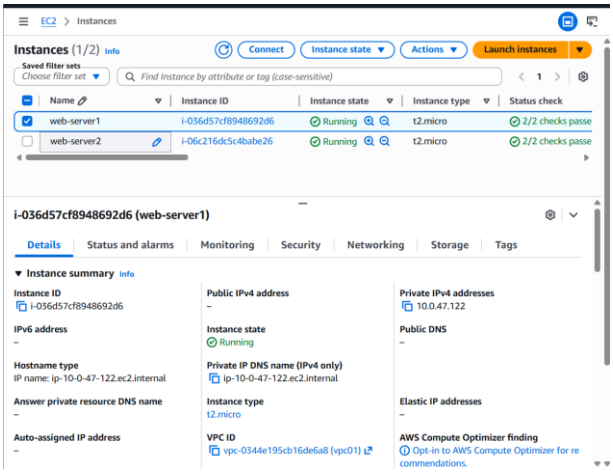
IAM Role



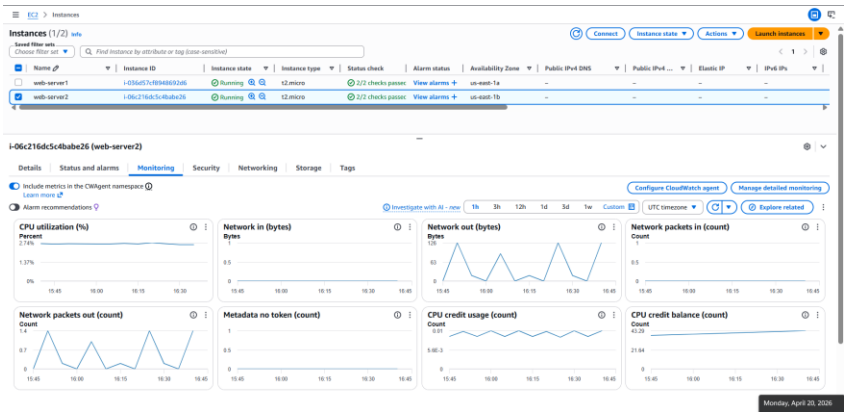
Secrets Manager



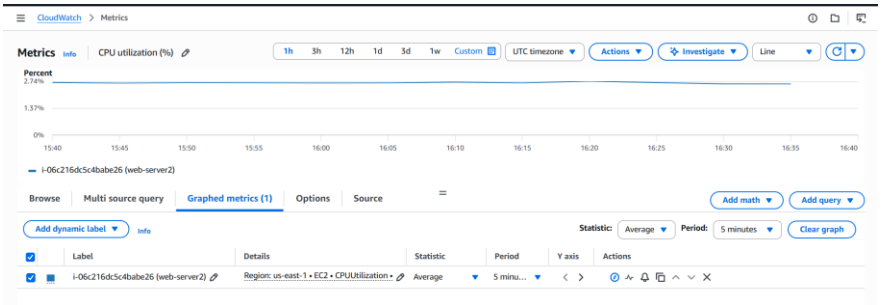
Least Privilege Principle



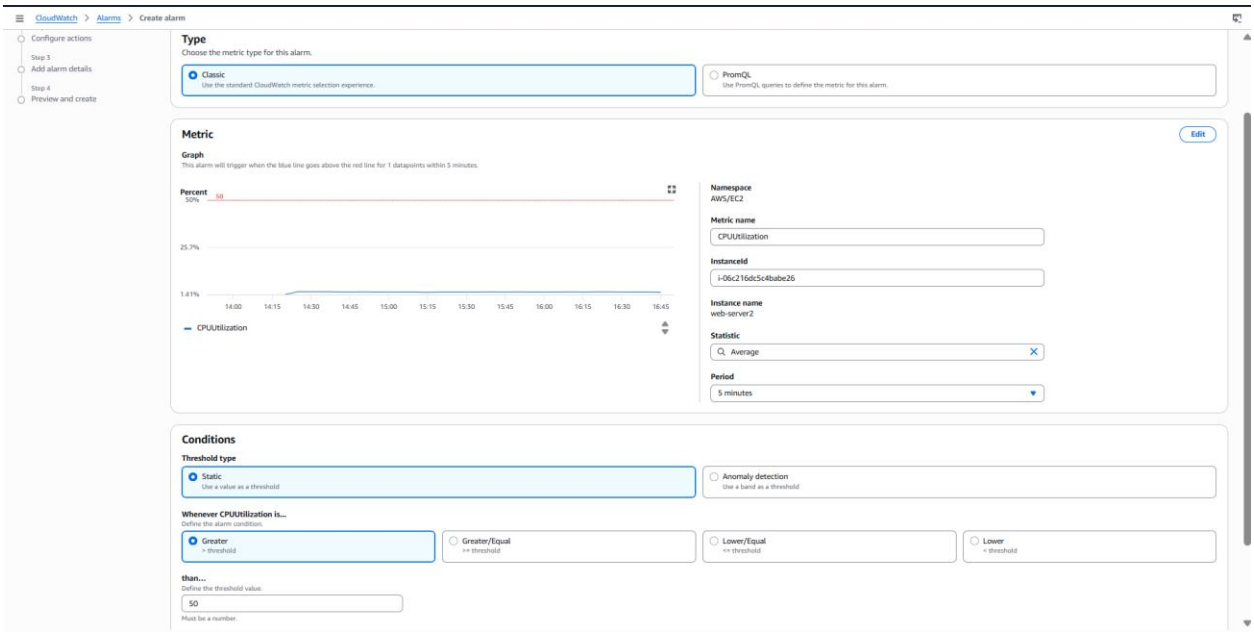
7. Monitoring & Logging

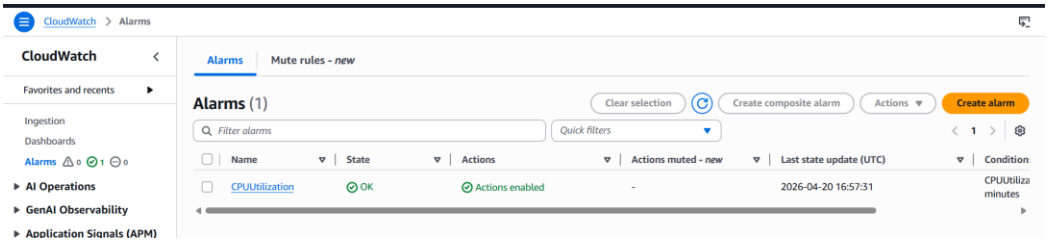


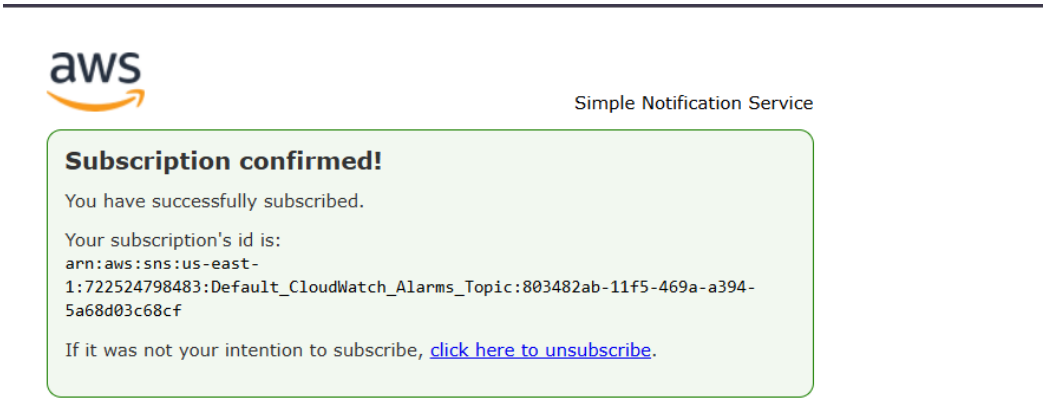
CloudWatch Metrics:

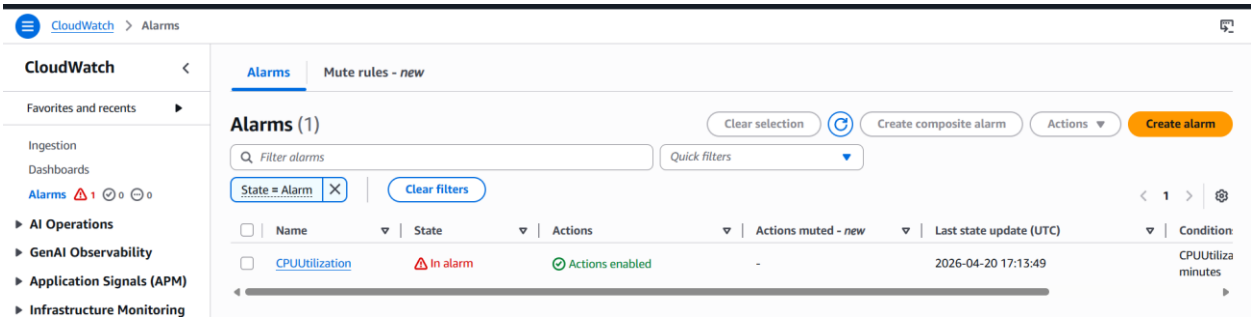


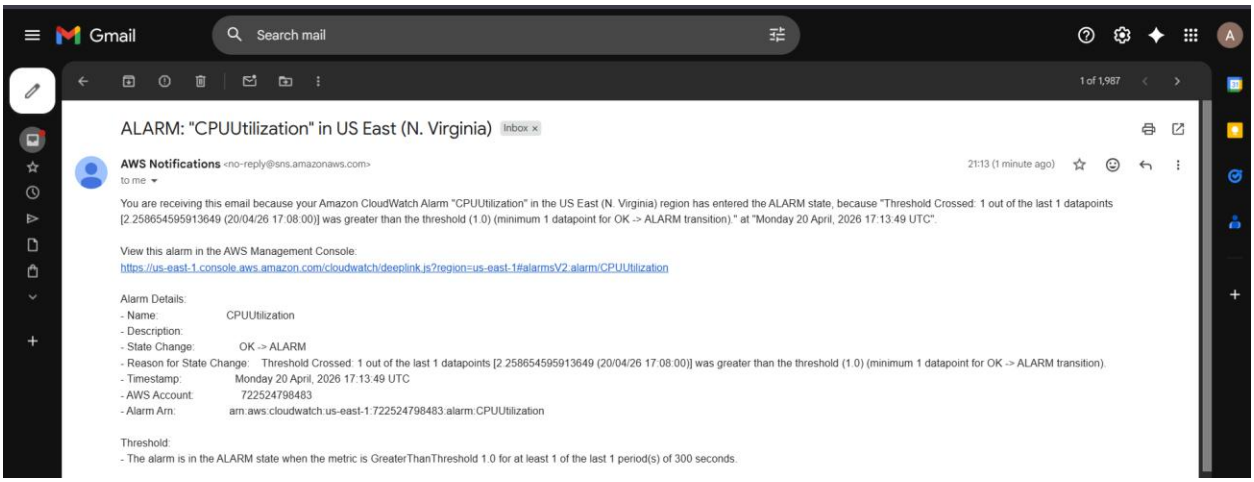
Alarm:


CloudWatch > Alarms > Create alarm
Configure actions
Step 3
Add alarm details
Step 4
Preview and create
Type
Choose the metric type for this alarm.
Classic
Use the standard CloudWatch metric selection experience.
PromQL
Use PromQL queries to define the metric for this alarm.
Metric
Edit
Graph
This alarm will trigger when the blue line goes above the red line for 1 datapoints within 5 minutes.
Percent
50%
50
25.7%
1.41%
14:00
14:15
14:30
14:45
15:00
15:15
15:30
15:45
16:00
16:15
16:30
16:45
CPUUtilization
Namespace
AWS/EC2
Metric name
CPUUtilization
InstanceId
i-06c216dc5c4babe26
Instance name
web-server2
Statistic
Average
Period
5 minutes
Conditions
Threshold type
Static
Use a value as a threshold
Anomaly detection
Use a band as a threshold
Whenever CPUUtilization is...
Define the alarm condition.
Greater
> threshold
Greater/Equal
>= threshold
Lower/Equal
<= threshold
Lower
< threshold
than...
Define the threshold value.
50
Must be a number.


CloudWatch > Alarms
CloudWatch
Favorites and recents
Ingestion
Dashboards
Alarms 0 1 0
AI Operations
GenAI Observability
Application Signals (APM)
Alarms
Mute rules - new
Alarms (1)
Clear selection
Create composite alarm
Actions
Create alarm
Filter alarms
Quick filters
1
Name
State
Actions
Actions muted - new
Last state update (UTC)
Condition
CPUUtilization
OK
Actions enabled
-
2026-04-20 16:57:31
CPUUtiliza
minutes

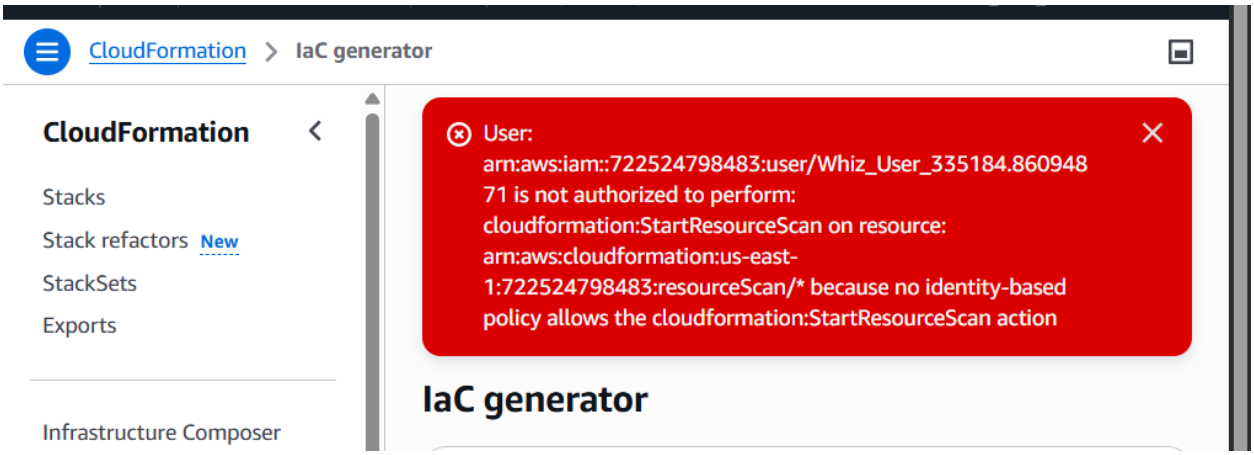

aws
Simple Notification Service
Subscription confirmed!
You have successfully subscribed.
Your subscription's id is:
arn:aws:sns:us-east-1:722524798483:Default_CloudWatch_Alarms_Topic:803482ab-11f5-469a-a394-5a68d03c68cf
If it was not your intention to subscribe, click here to unsubscribe.





I changed the alarm threshold to 1 just to trigger it quickly

8. Automation (Mandatory)




```
PS C:\Users\USER PC> terraform -version
Terraform v1.14.6
on windows_amd64
```

```
PS C:\Users\USER PC\terraform-file1> terraform init
Initializing the backend...
Initializing provider plugins...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v6.41.0...
- Installed hashicorp/aws v6.41.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
PS C:\Users\USER PC\terraform-file1>
```

```
PS C:\Users\USER PC\terraform-file1> terraform validate
Success! The configuration is valid.

PS C:\Users\USER PC\terraform-file1>
```